

## PSKEM, Uzbekistan

WMO: 384620

Lat: 41.9096N

Lon: 70.3671E

Elev: 1255

StdP: 87.13

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -14.0      | -11.4 | -20.3                         | 0.7 | -12.0 | -17.9 | 0.9 | -9.7 | 8.2                   | -3.5 | 7.4 | -3.5 | 0.1                   | N/A  | 0.472 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.9                   | 33.0            | 15.9 | 31.5 | 15.2 | 30.0 | 14.6 | 17.3                | 28.7 | 16.5 | 27.9 | 15.8 | 27.0 | 4.0                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 13.9                            | 11.5 | 20.0 | 12.8 | 10.7 | 18.9 | 11.9 | 10.1 | 18.5 | 53.8          | 28.9 | 51.1 | 27.9 | 48.9 | 27.0 | 24.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 8.2           | 7.1  | 5.7 | -17.0                      | 36.5 | 2.7                | 1.6 | -19.0                                              | 37.7 | -20.5      | 38.7 | -22.0      | 39.6 | -24.0      | 40.8 |    |    |
| (5)               |      |     | -17.4                      | 20.2 | 2.5                | 1.6 | -19.2                                              | 21.3 | -20.7      | 22.2 | -22.2      | 23.1 | -24.0      | 24.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.3   | -2.2  | -1.0  | 4.6   | 10.9  | 15.0  | 19.0  | 22.1  | 21.8  | 17.3  | 10.9  | 5.1   | -0.4  |  |  |
|                                                                        | DBStd       | 9.32   | 3.95  | 4.30  | 4.10  | 4.15  | 3.02  | 2.85  | 2.67  | 2.86  | 3.47  | 4.25  | 4.42  | 3.96  |  |  |
|                                                                        | HDD10.0     | 1420   | 379   | 309   | 172   | 38    | 3     | 0     | 0     | 0     | 1     | 40    | 154   | 323   |  |  |
|                                                                        | HDD18.3     | 3241   | 637   | 542   | 425   | 224   | 109   | 25    | 3     | 7     | 59    | 231   | 397   | 582   |  |  |
|                                                                        | CDD10.0     | 1536   | 0     | 0     | 6     | 64    | 158   | 271   | 374   | 366   | 219   | 69    | 8     | 0     |  |  |
|                                                                        | CDD18.3     | 313    | 0     | 0     | 0     | 1     | 5     | 46    | 118   | 114   | 27    | 1     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 4027   | 0     | 0     | 1     | 18    | 99    | 596   | 1472  | 1424  | 393   | 25    | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 1470   | 0     | 0     | 0     | 1     | 10    | 164   | 606   | 588   | 98    | 2     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 0.9    | 0.3   | 0.5   | 0.7   | 0.9   | 0.8   | 0.9   | 1.4   | 1.7   | 1.8   | 1.1   | 0.6   | 0.4   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 836    | 83    | 105   | 104   | 101   | 82    | 42    | 27    | 12    | 18    | 67    | 96    | 94    |  |  |
|                                                                        | PrecMax     | 1280   | 160   | 240   | 166   | 231   | 180   | 127   | 88    | 56    | 71    | 221   | 301   | 242   |  |  |
|                                                                        | PrecMin     | 463    | 22    | 14    | 27    | 17    | 10    | 2     | 0     | 0     | 0     | 3     | 13    | 17    |  |  |
|                                                                        | PrecStd     | 174    | 37    | 54    | 42    | 52    | 43    | 27    | 23    | 13    | 17    | 56    | 80    | 47    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 8.7   | 11.0  | 18.9  | 25.5  | 28.0  | 32.4  | 35.6  | 35.1  | 31.3  | 26.3  | 19.0  | 10.6  |  |  |
|                                                                        |             | MCWB   | 3.1   | 4.6   | 8.4   | 12.2  | 14.6  | 16.1  | 17.2  | 16.7  | 14.1  | 11.1  | 8.3   | 4.2   |  |  |
|                                                                        | 2%          | DB     | 6.1   | 8.2   | 15.6  | 23.0  | 25.9  | 30.1  | 33.4  | 32.9  | 29.2  | 23.1  | 16.0  | 7.9   |  |  |
|                                                                        |             | MCWB   | 1.6   | 3.0   | 7.1   | 11.4  | 14.0  | 15.4  | 16.3  | 15.6  | 13.1  | 10.2  | 6.9   | 2.8   |  |  |
|                                                                        | 5%          | DB     | 4.3   | 6.3   | 13.1  | 20.7  | 24.2  | 28.5  | 31.7  | 31.5  | 27.6  | 20.7  | 13.5  | 6.2   |  |  |
|                                                                        |             | MCWB   | 0.6   | 1.9   | 6.2   | 10.3  | 13.4  | 15.1  | 15.6  | 14.9  | 12.6  | 9.5   | 5.8   | 1.9   |  |  |
|                                                                        | 10%         | DB     | 2.8   | 4.6   | 10.7  | 18.3  | 22.4  | 26.8  | 30.0  | 30.1  | 25.6  | 18.4  | 11.3  | 4.7   |  |  |
|                                                                        |             | MCWB   | 0.1   | 1.0   | 5.1   | 9.5   | 12.8  | 14.8  | 14.9  | 14.3  | 11.7  | 8.7   | 5.0   | 1.2   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 3.9   | 5.6   | 9.5   | 14.2  | 16.4  | 18.1  | 19.2  | 17.8  | 15.3  | 12.6  | 10.1  | 5.5   |  |  |
|                                                                        |             | MCDB   | 7.3   | 10.1  | 15.0  | 20.8  | 24.4  | 26.4  | 29.4  | 30.8  | 27.3  | 21.0  | 15.8  | 8.2   |  |  |
|                                                                        | 2%          | WB     | 2.4   | 3.8   | 8.1   | 12.6  | 15.0  | 17.0  | 17.6  | 16.6  | 14.0  | 11.3  | 8.5   | 4.1   |  |  |
|                                                                        |             | MCDB   | 4.7   | 7.0   | 14.1  | 19.8  | 23.2  | 26.7  | 30.0  | 29.3  | 26.1  | 19.6  | 12.9  | 6.5   |  |  |
|                                                                        | 5%          | WB     | 1.4   | 2.5   | 6.9   | 11.4  | 14.2  | 16.2  | 16.6  | 15.9  | 13.2  | 10.4  | 7.1   | 2.7   |  |  |
|                                                                        |             | MCDB   | 3.4   | 5.3   | 11.9  | 18.3  | 22.3  | 25.7  | 28.9  | 28.6  | 24.9  | 18.3  | 11.7  | 5.3   |  |  |
|                                                                        | 10%         | WB     | 0.5   | 1.5   | 5.7   | 10.3  | 13.4  | 15.5  | 15.9  | 15.1  | 12.4  | 9.6   | 5.7   | 1.5   |  |  |
|                                                                        |             | MCDB   | 2.3   | 3.9   | 10.0  | 16.7  | 21.0  | 24.8  | 27.8  | 27.6  | 23.8  | 16.8  | 10.1  | 4.1   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.0   | 8.5   | 9.0   | 10.9  | 12.4  | 13.9  | 14.9  | 15.2  | 14.5  | 12.1  | 9.2   | 7.6   |  |  |
|                                                                        |             | MCDBR  | 9.1   | 10.1  | 11.6  | 14.7  | 15.2  | 16.2  | 16.7  | 16.9  | 16.7  | 15.4  | 12.6  | 9.2   |  |  |
|                                                                        | 5% WB       | MCWBR  | 6.3   | 6.7   | 6.3   | 6.8   | 7.0   | 6.3   | 5.4   | 5.5   | 5.8   | 6.4   | 6.5   | 6.0   |  |  |
|                                                                        |             | MCWBR  | 7.1   | 8.3   | 10.1  | 12.4  | 13.3  | 14.1  | 15.1  | 15.6  | 15.0  | 12.9  | 10.1  | 7.4   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.327 | 0.373 | 0.461 | 0.484 | 0.465 | 0.450 | 0.432 | 0.403 | 0.387 | 0.431 | 0.390 | 0.318 |  |  |
|                                                                        | taud        |        | 2.016 | 1.899 | 1.771 | 1.846 | 1.964 | 2.040 | 2.101 | 2.183 | 2.166 | 1.951 | 1.965 | 2.059 |  |  |
|                                                                        | Ebn at Noon |        | 777   | 796   | 765   | 787   | 817   | 829   | 840   | 854   | 838   | 724   | 687   | 751   |  |  |
|                                                                        | Edh at Noon |        | 115   | 153   | 198   | 197   | 180   | 167   | 156   | 139   | 131   | 144   | 122   | 101   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.97  | 2.84  | 3.82  | 4.89  | 6.24  | 7.20  | 7.38  | 6.73  | 5.37  | 3.25  | 1.97  | 1.56  |  |  |
|                                                                        | RadStd      |        | 0.23  | 0.36  | 0.40  | 0.58  | 0.46  | 0.40  | 0.38  | 0.32  | 0.28  | 0.46  | 0.32  | 0.20  |  |  |

## Historical Trends

|                             |       | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only           | DBAvg | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (47) Regional (9 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                             |       | N/A     | N/A    | N/A     | +0.57 | -0.44 | -0.73       | N/A     | N/A     | +96     |
|                             |       |         |        |         |       |       |             |         |         | +70     |

Nomenclature: See separate page